

CorpusMind & CorpusMind Lens

A User Guide for Linguists

Covers software version 1.0.0

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About This Guide

This guide is written for linguists — corpus linguists, discourse analysts, and multimodal-discourse researchers — who intend to use CorpusMind and its companion application, CorpusMind Lens, in their research or teaching. It assumes familiarity with the basic vocabulary of corpus linguistics (concordance, collocation, keyness, tagging) but not with software engineering, and it introduces the software's AI-assisted and vision-LM-powered features from first principles.

CorpusMind and CorpusMind Lens are two separate, independently installable desktop applications that share the same underlying analysis engine. CorpusMind is the primary application for text-corpus work; CorpusMind Lens is a focused companion for image-based and multimodal discourse analysis. Sections VI onward describe Lens specifically; everything before that concerns the shared engine and the primary CorpusMind application.

Every claim in this guide has been verified directly against the running software (version 1.0.0) rather than taken from marketing copy or release notes. Where a feature has a genuine, current limitation that could affect a research design — an incomplete translation, an export format that does not yet do what its label suggests, a content filter that is not fully comprehensive — this guide says so explicitly, in a shaded note, rather than omitting it. Researchers relying on this software for peer-reviewed work should treat these notes as part of their methods-section due diligence.

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I. Introduction and Design Philosophy

1. What CorpusMind Is

CorpusMind is a local-first, AI-assisted research environment for corpus linguistics. It combines a conventional corpus-analysis toolkit — concordance, frequency, collocation, keyness, dispersion, part-of-speech and dependency querying, discourse and sentiment analysis — with two things not typically found together in existing corpus software: a grounded AI research assistant that can query your corpus on your behalf and cite its evidence, and first-class support for Arabic alongside English, including CAMEL Tools-based morphological analysis, dialect identification, and full right-to-left interface support.

The application runs as a single background analysis engine ("the engine") with three interchangeable front ends: a desktop application, a browser-based progressive web app, and a self-hosted deployment. This guide describes the desktop application, which is the recommended mode for most linguists.

2. What CorpusMind Lens Is

CorpusMind Lens is a separate, separately installed desktop application for multimodal and visual discourse analysis. It shares CorpusMind's analysis engine but has its own interface, its own installer, and its own focus: images and image-text pairs rather than plain text corpora. Lens is best understood as a companion to CorpusMind, not a replacement — install it alongside CorpusMind when your research involves posters, advertisements, news photography, social-media imagery, or any material where meaning is constructed jointly by image and text.

Note — running both applications together

CorpusMind and CorpusMind Lens currently both expect to run their own copy of the analysis engine on the same local network port. If you start both applications at the same time, the second one to start may fail to launch its own engine and could behave unpredictably. Until this is resolved in a future release, the reliable workflow is to run one application at a time, or to close one before opening the other. This does not affect your data — each application still reads and writes to the same local project files.

3. Design Principles You Should Know

Nine principles govern how the software behaves, and understanding them will help you interpret its output correctly.

- Local-first, cloud-optional. Corpus text, images, and AI queries never leave your machine by default. Any cloud AI provider is an explicit, per-project opt-in, with a visible on-screen indicator whenever it is active.
- Grounded AI, never a bare chatbot. Every AI Assistant answer that makes an empirical claim carries a citation back to a concordance line, an image region, or a computed statistic that the engine can reproduce on demand. An answer with no such citation is visibly marked as ungrounded.
- Effect size and significance, always together. A word is never reported as "key" on the strength of a log-likelihood score alone; Log Ratio, %DIFF, Simple Maths, and Odds Ratio are reported alongside it, so you can judge practical as well as statistical significance.
- Zero-code, not zero-transparency. Every automatically generated result — a tag, a keyness list, a discourse-framework reading — is inspectable: you can see which model, formula, and software version produced it, and export that information as citeable methodology text.

- Interpretive claims are hypotheses, not facts. Anything belonging to the critical-discourse-analysis, framing, persuasion, or ideology family is labelled with the theoretical framework that produced it and phrased as a hypothesis ("under a [Framework] reading, X may indicate Y"), never as a bare assertion.
- Arabic is a first-class citizen. Dialect-aware tooling and the CAMEL Tools ecosystem are part of the core architecture, not an add-on — though, as noted in Part IV, interface translation coverage is still incomplete in this release.
- Practical scale, honestly stated. The current release is tested and reliable up to roughly 100,000 tokens per corpus on consumer hardware. Larger corpora will work but have not been systematically benchmarked; a scalable positional index for million-token corpora is on the roadmap, not yet implemented.
- Reproducibility is a feature. Every project records the exact tokenizer, tagger, and formula versions used, and can generate a ready-to-adapt Methods paragraph for a paper.
- Consent and restraint around biometric-adjacent features. Any feature that infers age, gender presentation, emotion, gaze, or similar attributes from a face is opt-in, off by default, and never performs identity recognition. See Part VI, Section 8 for the current scope and limitations of this safeguard.

4. Installation and System Requirements

CorpusMind and CorpusMind Lens are each distributed as native installers for Windows, macOS, and Linux, built from the same source and versioned together. Install them from the project's release page; the two applications do not share an installer.

- A local LLM runtime — Ollama or LM Studio — is required for the AI Assistant (CorpusMind) and for vision-LM features (Lens). Neither runtime is bundled; install one separately before first use.
- For CorpusMind's Arabic morphological analysis, the CAMEL Tools data package is required in addition to the base install.
- For CorpusMind Lens, at least one locally downloaded vision-capable model is required; the application can download one for you from within Settings (see Part VI, Section 4).

Note on hardware

Vision-language models are substantially heavier than the text models CorpusMind otherwise uses. The lightest supported option (moondream, roughly 1.8 billion parameters) runs on modest hardware; higher-fidelity options such as llama3.2-vision (11 billion parameters) benefit from 8 GB or more of available RAM. Choose the model that matches your machine before relying on Lens for a large image set.

II. Working with Corpora

1. Projects and Corpora

Work in CorpusMind is organised into projects. A project is a container for one or more corpora — your target corpus (or corpora) plus any reference corpora you attach to it — along with the AI Assistant conversations and exported results associated with that piece of research. Create a new project from the Home view before importing any text.

2. Building a Target Corpus

From Corpus Selection, choose Upload and add your source documents. Plain text, Word documents, and PDFs are supported; each uploaded document is automatically cleaned, segmented into sentences, tokenized, part-of-speech tagged, lemmatized, and dependency-parsed using spaCy for English or the CAMEL Tools pipeline for Arabic. This annotation pass is what every downstream analysis tool queries, so it is worth reviewing the resulting token and document counts before proceeding to analysis.

A corpus can be deleted or recompiled from Corpus Selection at any time. Recompiling re-runs the annotation pipeline — useful if you have upgraded the software and want a corpus re-tagged with a newer tagger version, or if you have added documents and want the corpus-level statistics refreshed.

3. Reference Corpora

Keyness analysis (Part III, Section 4) compares your target corpus against a reference corpus of general or comparandum language use. CorpusMind offers three ways to obtain one: download a small, redistributable frequency-list reference from within the app; download a larger, full-text reference corpus from its original academic host; or upload your own reference corpus, which is processed through the same annotation pipeline as a target corpus.

Bundled frequency-list references

These are small word-frequency profiles, redistributed with the application and verified by checksum on download, suitable for a first-pass keyness comparison:

Reference	Language	Size	License
BE06 — British English Written	English	Top 1,000 words	Open frequency data
Leipzig English News	English	Top 100 words	Leipzig Corpora Collection
Quranic Arabic	Arabic	Word-frequency list	Public domain text
CAMEL Arabic Frequency List	Arabic	Top 1,000 words	CAMEL Tools project

Full-text reference corpora

BNC Baby (a 4-million-word sample of the British National Corpus) and BAWE (British Academic Written English, 6.5 million words) are available as full-text downloads. These are academic corpora with restrictive licenses — see Part VII, Section 3 before using them in published work — and CorpusMind deliberately does not bundle their text with the software; downloading one retrieves it directly from its original academic archive at the time of use.

Uploading your own reference corpus

Use the Upload tab in the reference-corpus panel to supply your own comparison corpus — for instance, a diachronic baseline, a different genre, or a corpus from a prior study. It is processed through the identical annotation pipeline used for target corpora, and word frequencies are grouped case-insensitively for comparison, exactly as they are for the bundled references, so that keyness results are methodologically consistent regardless of which kind of reference you choose.

Note — minimum reference size

Keyness statistics such as log-likelihood become unstable with very small reference corpora. The software does not currently enforce a hard minimum, so exercise the same judgement you would with any keyness tool: a reference corpus of a few hundred words will produce numbers, but not reliable ones.

4. Corpus Versioning and Reproducibility

Each time a corpus is (re)compiled, CorpusMind records the exact annotation pipeline version used — tokenizer, tagger, and model identifiers. This information is available from the corpus detail view and is included automatically in the Methods-paragraph export (Part III, Section 14), so that a keyness or POS result you report in a paper can be reproduced, and its provenance checked, by a reviewer or a future version of yourself.

III. Core Analysis Tools

Once a corpus is loaded, the Analysis view exposes the following tools as tabs. Each computes over the annotated token stream produced during ingestion (Part II, Section 2), so the quality of tagging directly affects the quality of every result below.

Tool	What it does
Concordance (KWIC)	Search for a word, lemma, or pattern and browse it in key-word-in-context format, with adjustable context window and sortable columns.
Frequency	Word- or lemma-level frequency lists for the whole corpus or a defined subcorpus.
Collocation	Statistical collocates of a node word within an adjustable span, with a choice of association measures.
Keyness	Compares your target corpus against a reference corpus (Part II, Section 3) using log-likelihood alongside Log Ratio, %DIFF, Simple Maths, and Odds Ratio, per the effect-size principle in Part I.
Dispersion	How evenly a word or pattern is distributed across the corpus or across documents, useful for distinguishing a genuinely pervasive feature from one concentrated in a few texts.
N-grams	Frequent multi-word sequences of a specified length.
Part-of-Speech	Tag-level frequency and distribution, and POS-pattern search.
Grammar	Detects and quantifies passive constructions, modal verbs, negation, relative clauses, complex noun phrases, and tense — useful for register and stance studies.
Dependency	Query the dependency-parse structure directly, for syntactic patterns that POS tagging alone cannot capture.
Discourse	Metadiscourse analysis following Hyland's framework: hedges, boosters, self-mentions, engagement markers, and related categories.
Vocabulary	Lexical-profile statistics such as type-token ratio and vocabulary richness measures.
Sentiment	Corpus-level and document-level sentiment scoring.
Metaphor	Candidate metaphorical-usage detection, presented as candidates for manual review rather than a definitive classification, consistent with the hypothesis-framing principle in Part I.

14. Exporting Results

Every analysis panel offers an Export action. For Concordance, Frequency, Collocation, and Keyness, the chosen format — CSV, TSV, XLSX, or plain text — is produced correctly by the engine.

Note — export format for the remaining panels

For Dispersion, N-grams, Part-of-Speech, Grammar, Dependency, Discourse, Vocabulary, Sentiment, and Metaphor, the Export menu currently offers the same format choices, but every option other than JSON currently produces a JSON file regardless of the extension selected. If you need one of these results in a spreadsheet, export as JSON and convert it, or copy the on-screen table directly, until this is corrected in a future release.

IV. Arabic-Language Tools

Arabic support is implemented through CAMEL Tools and is available throughout the Analysis view for Arabic-language corpora, plus a dedicated Arabic Tools panel for the following, described here in the terminology used in Arabic computational-linguistics literature:

- Morphological analysis — root and pattern extraction, broken-plural detection, gender and number analysis.
- Buckwalter transliteration and diacritization/normalization.
- Dialect identification and register detection.
- Bilingual translation-equivalent lookup and parallel-corpus alignment and concordance.

Note — interface translation status

An Arabic interface language toggle exists (top bar), and switching it correctly changes the sidebar navigation labels and RTL layout direction. As of version 1.0.0, translation of the remaining screens — the analysis panels, the Arabic Tools panel itself, and most on-screen labels — has not yet been completed; most interface text remains in English regardless of the language toggle. This does not affect the accuracy of the Arabic NLP analysis itself, only the language of the surrounding interface. If your workflow depends on a fully Arabic interface, verify this against the version you have installed, as it is an active area of development.

V. The AI Research Assistant

1. Grounded AI: What "Grounded" Means Here

The Assistant is designed to answer questions about your loaded corpus by calling the same analysis tools described in Part III — running a real concordance search, a real keyness computation — and citing the result, rather than generating a plausible-sounding answer from general knowledge. An answer that could not be tied to a specific tool call is visibly marked as ungrounded, so you always know whether a claim is backed by your data or by the model's general training.

2. Setting Up a Local Model

Install Ollama or LM Studio (Part I, Section 4), pull or download a model, and select it in Settings → Model Providers. The Assistant works entirely offline once a local model is installed; no cloud account is required.

3. Using the Assistant

The Assistant sidebar offers a persistent set of suggested questions, which combine fixed starting points with questions generated dynamically from your loaded corpus's own statistics once one is active — for example, a suggestion to compare the corpus's strongest keywords against its reference corpus. These remain available throughout a conversation, not only before your first message.

4. Reading Assistant Output

A grounded answer includes a citation back to the concordance line, statistic, or computed result that supports it, so you can verify the claim directly in the relevant analysis panel. Treat an ungrounded answer as you would any general-purpose language-model output: plausible, but not evidence.

5. Limitations

Answer quality depends substantially on the local model you have chosen; smaller models are faster but less reliable at correctly invoking analysis tools. If the Assistant's answers seem generic or fail to cite a specific result, try a larger local model before concluding the feature does not work.

VI. CorpusMind Lens: Multimodal and Visual Discourse Analysis

CorpusMind Lens (Part I, Section 2) applies the same grounded, framework-lensed, hypothesis-framed philosophy described in Part I to images and image-text pairs. Everything in this part concerns the Lens application specifically.

1. Installing Lens Separately

Lens has its own installer and its own application window, distinguished visually by a blue interface theme (CorpusMind's is green) so that the two are never mistaken for one another at a glance.

2. Image Sets

Create an image set — the multimodal equivalent of a corpus — and upload images, optionally paired with co-occurring text such as a caption, headline, or the article body the image accompanied. This paired text feeds directly into the discourse-framework analyses in Section 5 below.

3. Automatic Image Analysis

On upload, every image is analysed immediately using OCR text extraction, colour-palette statistics, and a composition analysis based on Kress and van Leeuwen's (2006) visual-grammar framework — information value, salience, and framing. This pass requires no AI model and completes instantly.

4. Vision-LM Description and Downloading a Model

With a local vision-capable model installed, the Describe action produces a grounded natural-language description of the image, cross-checked against the OCR pass rather than silently overwriting it when the two disagree. Download a vision model from Settings → Model Providers:

Model	Size	Notes
moondream	1.7 GB (1.8B parameters)	Runs on modest hardware; a reasonable default for basic description.
llama3.2-vision:11b	7.9 GB (11B parameters)	Higher-fidelity description and discourse analysis; needs 8 GB+ RAM.
gemma3:4b	3.3 GB (4B parameters)	Multilingual, including Arabic image content.

5. The Eight Discourse-Analysis Frameworks

Every image can be read through eight discourse-analytic lenses, selectable individually or in combination. Each is clearly labelled by framework and phrased as a hypothesis, per the interpretive-claims principle in Part I:

Framework	Theoretical basis
Social Semiotic	Kress and van Leeuwen's grammar of visual design
Critical Discourse Analysis	Selectable sub-framework: Fairclough, van Dijk, Wodak, or Machin & Mayr
Persuasion	Classical rhetoric (Aristotle) and Toulmin's argument structure

Framework	Theoretical basis
Framing	Entman's framing functions
Narrative	Labov's narrative-structure schema
Visual Metaphor	MIPVU-inspired cross-modal metaphor identification
Emotion	Combined image- and text-based affective analysis
Cultural / Symbolic	Culturally situated symbolic reading

Each route can run in heuristic mode (fast, based on the colour/composition/OCR/caption data from Section 3, no model required) or vision-LM mode (richer, model-generated, appended with `?mode=llm`). If the vision-LM call fails for any reason, the route falls back to heuristic mode automatically rather than returning an error.

Note — what the heuristic mode actually sees

Without a vision-LM and without a caption you provide at upload, the heuristic-mode discourse readings are computed from colour statistics, geometric composition, and OCR text alone — the software has no independent way to know what the image depicts. For any framework whose reading depends on subject matter, treat heuristic-mode output as a much thinner signal than vision-LM-mode output, and prefer the latter whenever a vision model is available.

6. Image–Text Alignment

The alignment inspector shows which region of an image is claimed to correspond to which span of the paired text, in both a heuristic and a vision-LM mode, viewable side by side so you can judge which is more reliable for your material.

7. Batch View

Once several images in a set have cached analysis, the batch view aggregates recurring discourse themes across the set and produces an OCR-derived word-frequency list for the whole set — useful for a first-pass overview before close reading individual images.

8. Facial Analysis and the Consent Gate

Per the consent-and-restraint principle in Part I, any output that describes a person's estimated age group, gender presentation, facial expression, or physical appearance is, by default, withheld: person-descriptive language is detected and replaced with a redaction marker before it reaches the interface, across the Describe route and all eight discourse frameworks, including on cached results. The gate can be opened explicitly in Settings if your research specifically requires this category of output and your ethics approval covers it.

Note — current scope of the filter, read before relying on it for sensitive material

The filter works by scanning generated text for a defined vocabulary of age, gender, expression, and physical-appearance terms. Verified directly against this release: it reliably catches the common case (a vision-LM volunteering age or gender commentary unprompted), but it does not currently catch every category of person-descriptive content a vision-LM may produce — in direct testing, statements referencing a person's apparent ethnicity, religious or cultural dress, or speculative socioeconomic status passed through unfiltered. If your research involves imagery where any of these categories could appear in model output — for instance, studies of media representation involving identifiable individuals — do not rely on this filter as a complete safeguard.

Review vision-LM output manually before it is used, quoted, or stored in a way your ethics protocol would classify as processing that category of data, regardless of the gate's setting.

VII. Data Handling, Privacy, and Licensing

1. Local-First Philosophy

By default, no corpus text, no image, and no AI query leaves your machine. Both applications run their analysis engine and any language model locally; a cloud AI provider is available only as an explicit, visibly indicated, per-project opt-in.

2. What Never Leaves Your Device

Uploaded documents and images, their annotations, concordance and keyness results, and Assistant conversations are stored locally and are not transmitted anywhere unless you explicitly enable a cloud provider or manually export and share a file yourself.

3. Licensing of Reference Corpora

The small, bundled frequency-list references in Part II, Section 3 are redistributable and shipped with the software. BNC Baby and BAWE are not: both are distributed under the British National Corpus family of end-user licences, which forbid redistribution of the corpus text and describe an individual approval process for access. CorpusMind accordingly does not bundle their text; downloading either retrieves it directly from its academic host at the time of use, under the terms that host sets. If you intend to publish results derived from either corpus, cite them according to their own licence terms, not as a CorpusMind-distributed resource.

4. Citing CorpusMind in Your Research

Mandour, W., & Ibrahim, W. (2026). *CorpusMind: A local-first, AI-native research environment for corpus linguistics and multimodal discourse analysis* (Version 1.0.0) [Computer software]. Zenodo.
<https://doi.org/10.5281/zenodo.21226650>

Waleed Mandour, Sultan Qaboos University (ORCID 0000-0002-9262-5993); Wesam Ibrahim, Princess Nourah bint Abdulrahman University (ORCID 0000-0003-0710-6038). A BibTeX entry is available in the project's repository README.

VIII. Troubleshooting

Symptom	What to try
Engine will not start / "engine offline"	Check Settings → System for a diagnostic log. If you have both CorpusMind and CorpusMind Lens open at once, close one — see the note in Part I, Section 2.
AI Assistant gives generic, ungrounded answers	Confirm a local model is selected in Settings → Model Providers and that Ollama or LM Studio is running. Try a larger model if answers remain generic.
Downloading a bundled reference corpus (e.g. BNC Baby) fails	Confirm your internet connection can reach the academic host; the file is fetched at download time, not bundled. Retry — this has been a source of fixed defects in earlier releases.
An exported file will not open correctly in Excel	See the export-format note in Part III, Section 14 — several panels currently export JSON regardless of the extension chosen.
Vision-LM features in Lens are unavailable	Download a vision-capable model (Part VI, Section 4) — the heuristic-only analysis works without one, but Describe and the LLM discourse modes require it.

Appendix — Glossary

Term	Definition
Concordance / KWIC	A list of every occurrence of a search term shown with surrounding context, aligned in columns (key-word-in-context).
Keyness	A statistical comparison of word frequencies between a target corpus and a reference corpus, identifying words that are unusually frequent (or infrequent) in the target.
Log-likelihood, Log Ratio, %DIFF, Simple Maths, Odds Ratio	Statistical measures used together in keyness analysis: the first for statistical significance, the remainder for effect size (practical significance).
Grounded (AI) answer	An AI Assistant response that cites a specific, reproducible result from your own corpus, as opposed to a general, unverifiable claim.
Heuristic mode (Lens)	Image analysis computed from colour, composition, OCR text, and caption alone, without a vision-language model.
Vision-LM mode (Lens)	Image analysis performed by a locally run vision-language model, producing a grounded natural-language reading of the image itself.
Consent gate	The opt-in safeguard restricting person-descriptive output (age, gender presentation, expression, appearance) from vision-LM analysis; see Part VI, Section 8 for its current scope.
Corpus versioning	CorpusMind's record of the exact tokenizer, tagger, and model versions used to annotate a given corpus, enabling reproducible reporting.